BEYER-GARRATT

There is something undeniably impressive about the monsters known, quite lovingly, as the Garratts. It could be that they are among the heaviest and biggest steam locomotives ever built. It could also be because of the immense power they wielded as they hauled heavy loads, often up steep gradients. Quick and efficient, these giants could run at high speeds without causing too much damage to the tracks – specially the lightly laid ones. They were also multipurpose and could be used for passenger trains or freight.

The Garratts employed the kind of power that was quite unlike the steam locomotives of their time. They were known for their ability to, quite easily, haul a 2,400-tonne train on a 1 in 100 gradient at about 72 km/h. The articulated locomotives had three separate frames – unlike the conventional steam ones where the whole unit is carried on a single frame. This articulation allowed it to seamlessly negotiate tight curves. This was why the Garratts proved highly effective for the Darjeeling Himalayan Railway (DHR), where they were first used in 1909. The locomotives almost brought about a revolution in steam traction with these innovations.

The Beyer-Garratt had a single boiler flanked by two tenders at either end with a reciprocating engine for each tender. The power units carried the water tanks while the rear unit also shouldered the fuel supply. The boiler and firebox unit were slung between the two engine units which freed them of size constraints. The large firebox allowed for efficient fuel combustion as well, thereby increasing the transfer of heat to the boiler and helping the Garratt belt out its legendary power and tremendous performance while belching little smoke.

Garratts ran on the Anara–Bhojudih (in West Bengal) and Bhilai–Dalli Rajhara (in modern-day Chhattisgarh) sections of the Bengal–Nagpur Railway (BNR) till 1970. They also hauled iron ore in the gradient section of Dangoaposi and Tatanagar in BNR's Chakradharpur division in present-day Jharkhand. As technology improved, the engines